

How available information shapes denoising and computation

I: participants

woman, man, blue cup

R: interaction roles

giver, receiver, object

V: visual realization

hand and cup positions

A. What does clearer role information contribute to visual prediction?

Hold participants, visual-target noise, and transformer weights fixed.

Noisier roles R

woman ← who gives? → man

Denoise the same hand/cup states V

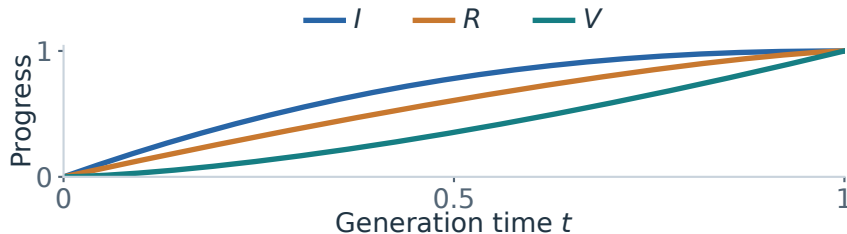
Clearer roles R

woman — gives cup —> man

Denoise the same hand/cup states V

Compare visual denoising losses; role labels are withheld from the prompt.

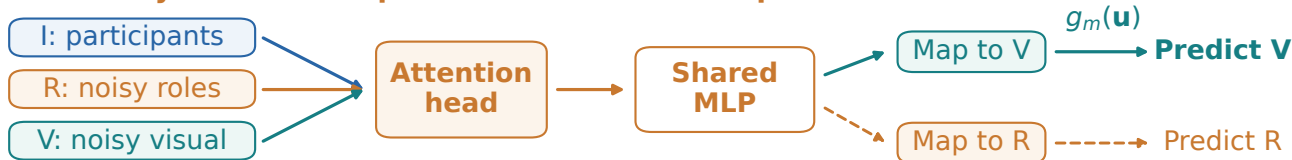
B. Learn component schedules with the transformer



Earlier R can help V,
but R must be
predicted with less
visual information.

All components
continue developing.

C. Identify information paths and reusable computations



Gate $g_m(\mathbf{u})$: noise-dependent contribution. Dashed: reuse for role prediction.